Richard Álvarez

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About Me

I am a software engineer and researcher with a background in machine learning, data-driven storytelling, and computational media. My work combines programming, digital humanities, and film to develop tools that enhance creative workflows. I have authored two research papers on AI-driven media analysis, developed over 17 open-source projects, and built production-level software for data processing and recommendation systems.

Education

New York University (NYU) Tandon School of Engineering — Brooklyn, New York
Master of Science in Computer Engineering — *Aug 2025*

Kenyon College — Gambier, Ohio
Bachelor of Arts in Film; GPA: 3.4/4.0 — *Aug 2020 – May 2024*
Minor in History and Concentration in Integrated Program in Humane Studies

Authored two papers on machine learning applications in creative industries. Produced and edited over 15 experimental video projects, including music videos, audio-reactive visualizations, and short films.

Relevant coursework includes Senior Research Seminar (IPHS 484), AI for the Humanities (IPHS 300), Advanced Post-Production (FILM 391), Data Structures and Program Design (SCMP 218), Digital Photography (ARTS 321), Sex, Drugs, Guns: Research Strategies in the Contemporary Age (INDS 140), and Software Development (SCMP 318).

Work Experience

IT Assistant — Gambier, Ohio
Library and Information Services (LBIS), Kenyon College · Part-time — *Sep 2023 – Feb 2024*
I supported campus-wide technology needs by preparing workstations, moving office tech, and securely erasing and recycling equipment. I restocked printers daily. I conducted classroom checks under the guidance of team members. I streamlined team projects by applying programming skills, in one instance by generating a spreadsheet of course meetings and classroom locations to determine when our techs could perform maintenance.

Video Editor — Gambier, Ohio
Kenyon College · Contract — *Apr 2022 – May 2022*
I condensed over 30 hours of interviews into a concise 10-minute recap for the John W. Adams Summer Scholars Program in Socio-Legal Studies. I crafted a polished visual presentation that adhered to Kenyon's Visual Identity System. I highlighted key interview themes while ensuring balanced representation of all participants. I delivered the final product on schedule in an optimized format, maintaining professional collaboration with the employer throughout the project.

Research Assistant — Chicago, Illinois
University of Chicago · Part-time — *Aug 2018 - Nov 2019*
Worked under Bernard Dickens III on strategies to protect against supply-chain attacks and ensure file integrity using advanced checksum technologies. Attended monthly code reviews, contributed 27 commits, and developed efficient data integrity verification scripts. Gained experience with data processing pipelines and research-driven engineering.

Publications

Unsupervised Deep Learning and PySceneDetect Analysis | *GitHub* | *Digital Kenyon* | May 23rd 2023
This research focused on analyzing short-format video editing trends by leveraging PySceneDetect and unsupervised deep neural networks. Advanced data visualization techniques, including t-SNE and PCA, were employed to uncover patterns and gain insights into the editing styles and trends prevalent in the dataset.

Certificates

CompTIA ITF+ *Sep 2024*
NYU Tandon Bridge *Mar 2025*

Skills

Programming: Python, JavaScript/TypeScript, SQL, C/C++
Frameworks: Next.js, React, Tailwind CSS, Scikit-Learn, Keras, pandas, NumPy, Databricks, PySpark
Machine Learning: Diffusion models, LoRA, LxMs, retrieval-augmented generation (RAG), fine-tuning LLMs
Technical Workflows: Agile, CI/CD, DevOps, OOP, functional programming
Systems & Tools: Linsux, shell scripting, AWS, DigitalOcean

Websites

Machine Television Visit Site
Online Store *Oct 2024*
Developed a functional e-commerce platform for an independent skate brand. The site was built using Next.js and Tailwind CSS for an intuitive front-end, paired with Node.js for a robust back-end infrastructure. Integrated Stripe API for seamless payment processing, optimizing user workflows across desktop and mobile.

Joaquin Morales Visit Site
Portfolio *Jan 2025*
Designed and deployed a dynamic portfolio site for a professional cinematographer. The project used Next.js for high performance, with Tailwind CSS for responsive design. Implemented a custom CMS to enable efficient content updates, managing galleries and testimonials with ease. Leveraged DigitalOcean S3 storage for scalability and fast load times for video and photo content.

GREasyVocab Flashcards
Web App *Jul 2024*
Created a personalized GRE vocabulary tool powered by OpenAI's APIs. The application leverages LangChain to provide personalized prompts tailored to user inputted data. Developed a secure full-stack system with user authentication and database management, ensuring a smooth and customized learning experience.

Additional Projects

A Retrieval-Augmented Film Recommendation System | *GitHub* | *Digital Kenyon* | May 8th 2024
This project utilized LangChain's OpenAI integration to dynamically generate queries based on user preferences, showcasing the potential of advanced AI and machine learning in digital entertainment. The Retrieval-Augmented Film Recommendation System was developed using Node.js and integrated with the OMDb and TMDb APIs to enhance movie metadata, delivering precise and personalized recommendations.

AI-Driven Kubrick-Inspired Film Script Generation | *GitHub*
Designed and developed an AI pipeline to generate film scripts inspired by Stanley Kubrick's cinematic style. Leveraging Dust.tt and Large Language Models, I created a custom API to enable dynamic and stylistically consistent script generation. This project demonstrated the potential of generative AI for creative industries, producing scripts that emulated Kubrick's distinctive narrative and thematic characteristics.

Sentiment Analysis of Rotten Tomatoes Reviews | *GitHub*
Conducted a sentiment analysis of user reviews from Rotten Tomatoes to evaluate the psychology of movie consumers and the reliability of the platform's 'freshness' indicators. Using VADER and the NLTK library, I processed text data by filtering stop words, tokenizing reviews, and extracting sentiment scores. The analysis revealed that 73 of user reviews aligned with their assigned 'freshness' labels, validating both the reliability of Rotten Tomatoes user ratings and the effectiveness of the VADER sentiment analysis tool. Findings were presented in detailed visual reports, highlighting correlations between sentiment and user ratings.